



## **Data Analysis & Business Understanding**

### **AI/ML Internship**

#### **Week 1 Task**

**Title:** Exploratory Data Analysis of the Online Retail II Dataset

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## Project Overview

This project performs Exploratory Data Analysis (EDA) on the Online Retail II dataset to understand customer purchasing behavior. The analysis identifies sales trends, top-selling products, country-wise performance, and data quality issues using Python and visualization libraries.

## Objective

The objective that this analysis aims to achieve includes the following:

- Analyse the Online Retail II dataset to understand customer purchasing behavior.
- Examine the dataset for missing values, duplicate records, and overall data quality.
- Identify the top-selling products based on total quantity sold and revenue generated.
- Evaluate sales performance across different countries to determine key markets.
- Analyse monthly revenue trends to identify sales patterns over time.
- Explore relationships between numerical variables using correlation analysis.
- Detect potential outliers in the dataset using box plots.
- Derive meaningful business insights to support future data-driven decision-making and recommendation systems.

## Dataset Information

### Dataset Name:

Online Retail II

### Source:

UCI Machine Learning Repository

### Description:

The dataset contains transactions from a UK-based online retail company between 2009 and 2011. Each record represents one purchased product in a customer invoice. The dataset is widely used for exploratory data analysis, customer behavior analysis, and machine learning applications.

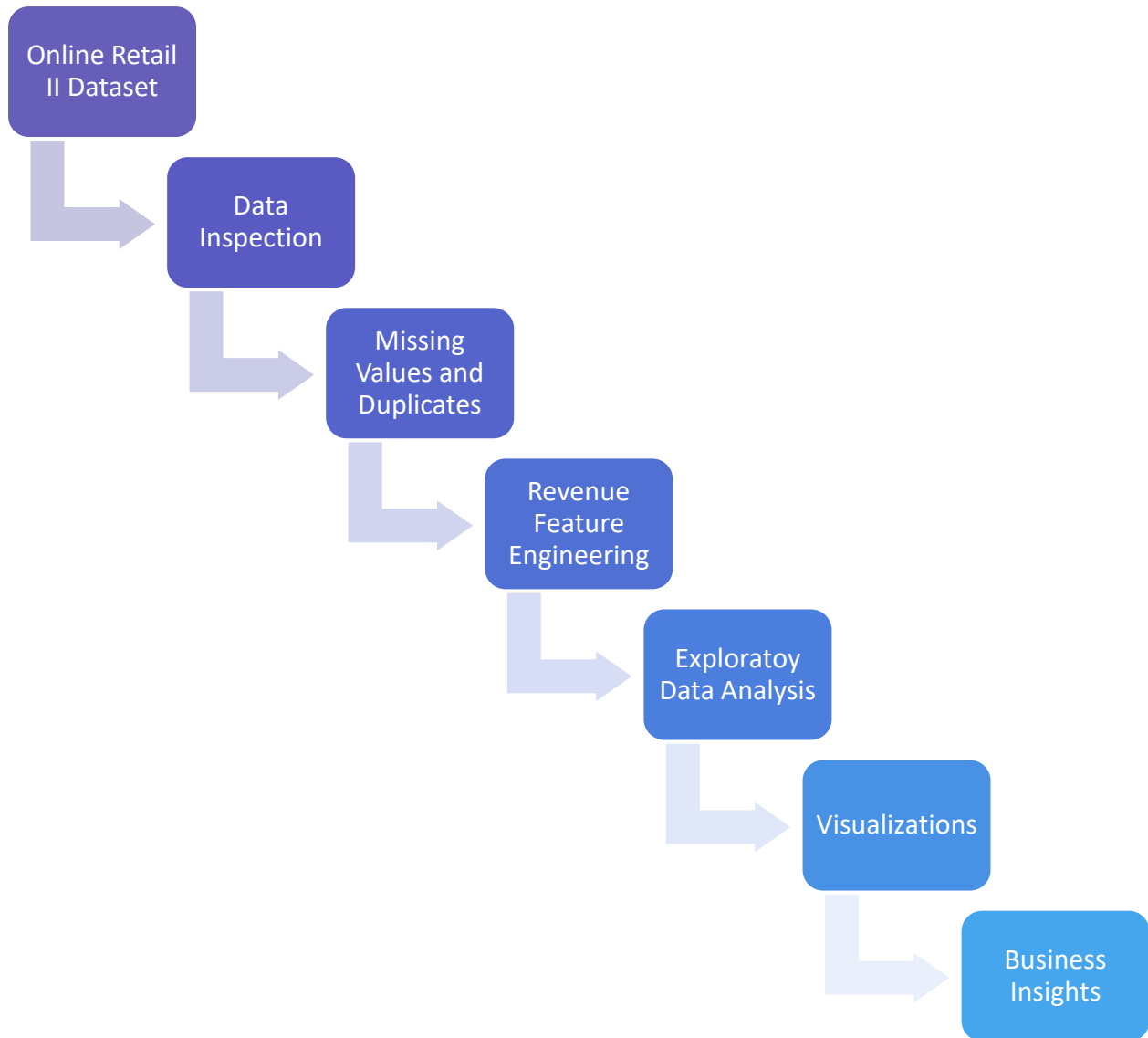
## Dataset Features:

| Feature     | Description                                |
|-------------|--------------------------------------------|
| Invoice     | Unique invoice number for each transaction |
| StockCode   | Unique product code                        |
| Description | Name of the purchased product              |
| Quantity    | Number of items purchased                  |
| InvoiceDate | Date and time of the transaction           |
| Price       | Unit price of the product                  |
| Customer ID | Unique identifier for each customer        |
| Country     | Customer's country                         |

## Tools Used

| Tool             | Purpose                   |
|------------------|---------------------------|
| Python           | Programming Language      |
| Jupyter Notebook | Code Execution            |
| Pandas           | Data Manipulation         |
| NumPy            | Numerical Computations    |
| Matplotlib       | Data Visualization        |
| Seaborn          | Statistical Visualization |

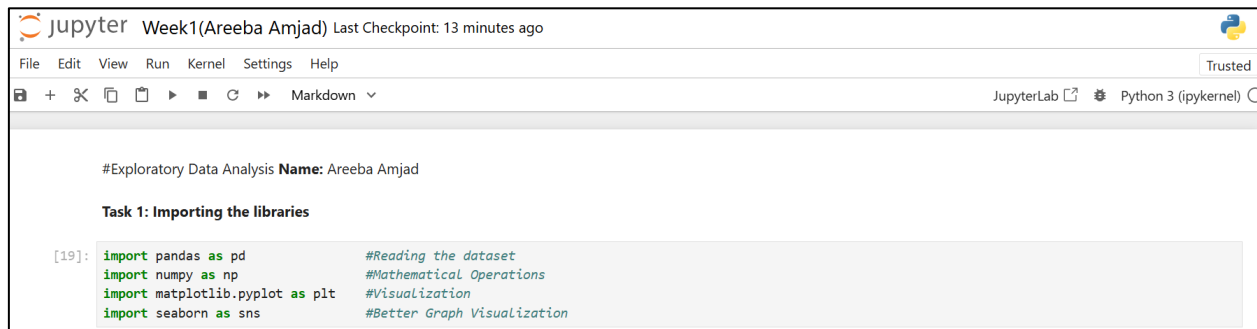
## Work Flow Diagram



# Task Implementation

## Importing the Libraries

All the necessary libraries are imported once so they can be used later on without importing them again and again.



The image shows a JupyterLab window titled "Week1(Areeba Amjad) Last Checkpoint: 13 minutes ago". The interface includes a menu bar (File, Edit, View, Run, Kernel, Settings, Help) and a toolbar with icons for file operations and execution. The main area displays a code cell for "Task 1: Importing the libraries". The code imports pandas, numpy, matplotlib.pyplot, and seaborn, each with a comment explaining its purpose.

```
#Exploratory Data Analysis Name: Areeba Amjad

Task 1: Importing the libraries

[19]: import pandas as pd          #Reading the dataset
      import numpy as np          #Mathematical Operations
      import matplotlib.pyplot as plt #Visualization
      import seaborn as sns       #Better Graph Visualization
```

## Loading the Dataset



The image shows a JupyterLab window titled "Task 2: Loading the Retail Dataset". It displays a code cell where the dataset is loaded from an Excel file using pandas.

```
Task 2: Loading the Retail Dataset

[9]: ds=pd.read_excel("C:/Users/Laptop Zone/OneDrive/Desktop/online_retail_II.xlsx")
```

## Displaying Basic Information

In order to understand the basic structure of the data its shape, columns and data type were examined.



The image shows a JupyterLab window titled "Task 3: Displaying Basic Information". It displays a code cell where the dataset's shape, columns, and data types are printed. The output shows the dataset has 525461 rows and 8 columns. The columns are Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, and Country. The data types are object, object, object, int64, datetime64[us], float64, float64, and str respectively.

```
Task 3: Displaying Basic Information

[10]: print("\n=====Shape of Dataset:=====\n",ds.shape)
      print("\n=====Columns of Dataset:=====\n")
      print(ds.columns)
      print("\n=====DataType of Dataset:=====\n")
      print(ds.dtypes)

=====Shape of Dataset:=====
(525461, 8)

=====Columns of Dataset:=====
Index(['Invoice', 'StockCode', 'Description', 'Quantity', 'InvoiceDate',
       'Price', 'Customer ID', 'Country'],
      dtype='str')

=====DataType of Dataset:=====
Invoice           object
StockCode         object
Description        object
Quantity          int64
InvoiceDate       datetime64[us]
Price             float64
Customer ID       float64
Country           str
dtype: object
```

Figure 1: Dataset Overview

## Missing Values and Duplicate Rows

The quality of data was examined by checking the columns having null values and duplicate records. Data was only observed, no deletion of data is performed.

```
Missing Values:

Invoice          0
StockCode        0
Description      2928
Quantity         0
InvoiceDate      0
Price            0
Customer ID     107927
Country          0
dtype: int64

Duplicated Rows: 6865
```

Figure 2: Missing Values and Duplicate Records

## Revenue Feature Engineering

A new column was added to the data set with the name revenue which was the product of Quantity and price. Later top 5 rows from the data were displayed.

```
Revenue Calculation

[12]: ds["Revenue"] = ds["Quantity"] * ds["Price"] #Creating a new column of Revenue in Dataset
      ds.head() #Displaying few rows from Dataset

[12]:
```

|   | Invoice | StockCode | Description                         | Quantity | InvoiceDate         | Price | Customer ID | Country        | Revenue |
|---|---------|-----------|-------------------------------------|----------|---------------------|-------|-------------|----------------|---------|
| 0 | 489434  | 85048     | 15CM CHRISTMAS GLASS BALL 20 LIGHTS | 12       | 2009-12-01 07:45:00 | 6.95  | 13085.0     | United Kingdom | 83.4    |
| 1 | 489434  | 79323P    | PINK CHERRY LIGHTS                  | 12       | 2009-12-01 07:45:00 | 6.75  | 13085.0     | United Kingdom | 81.0    |
| 2 | 489434  | 79323W    | WHITE CHERRY LIGHTS                 | 12       | 2009-12-01 07:45:00 | 6.75  | 13085.0     | United Kingdom | 81.0    |
| 3 | 489434  | 22041     | RECORD FRAME 7" SINGLE SIZE         | 48       | 2009-12-01 07:45:00 | 2.10  | 13085.0     | United Kingdom | 100.8   |
| 4 | 489434  | 21232     | STRAWBERRY CERAMIC TRINKET BOX      | 24       | 2009-12-01 07:45:00 | 1.25  | 13085.0     | United Kingdom | 30.0    |

## Top 10 best-selling products

Using the Description column the products were grouped. The top 10 products were identified based on total Quantity sold and total Revenue generated.

### ❖ Quantity:

```
[13]: #Displaying Top 10 products grouped by Quantity
      top_Quantity = ds.groupby("Description")["Quantity"].sum()
      top_Quantity = top_Quantity.sort_values(ascending=False)
      top_Quantity = top_Quantity.head(10)
      print("\n===== Top 10 Products by Quantity =====\n\n", top_Quantity)
```

```
===== Top 10 Products by Quantity =====

Description
WHITE HANGING HEART T-LIGHT HOLDER    57733
WORLD WAR 2 GLIDERS ASSTD DESIGNS     54698
BROCADE RING PURSE                     47647
PACK OF 72 RETRO SPOT CAKE CASES      46106
ASSORTED COLOUR BIRD ORNAMENT          44925
60 TEATIME FAIRY CAKE CASES           36326
PACK OF 60 PINK PAISLEY CAKE CASES    31822
JUMBO BAG RED RETROSPOT               30727
SMALL POPCORN HOLDER                  29500
STRAWBERRY CERAMIC TRINKET BOX         26563
Name: Quantity, dtype: int64
```

Figure 3: Top 10 Products by Quantity

## ❖ Revenue:

```
#Displaying Top 10 products grouped by Revenue
top_Revenue= ds.groupby("Description")["Revenue"].sum()
top_Revenue = top_Revenue.sort_values(ascending=False)
top_Revenue = top_Revenue.head(10)
print("\n===== Top 10 Products by Revenue=====\\n\\n",top_Revenue)
```

===== Top 10 Products by Revenue=====

| Description                         |           |
|-------------------------------------|-----------|
| REGENCY CAKESTAND 3 TIER            | 163051.46 |
| WHITE HANGING HEART T-LIGHT HOLDER  | 157865.43 |
| DOTCOM POSTAGE                      | 116401.99 |
| ASSORTED COLOUR BIRD ORNAMENT       | 72454.12  |
| PAPER CHAIN KIT 50'S CHRISTMAS      | 57870.20  |
| JUMBO BAG RED RETROSPOT             | 56444.61  |
| PARTY BUNTING                       | 49645.52  |
| ROTATING SILVER ANGELS T-LIGHT HLDR | 47672.49  |
| POSTAGE                             | 46092.36  |
| JUMBO BAG STRAWBERRY                | 36699.61  |

Name: Revenue, dtype: float64

Figure 4: Top Products by Revenue

## Sales performance analysis by country

Country-wise analysis was performed in order to calculate the total revenue, total quantity sold and total number of unique orders placed.

### Task 7: Sales Performance by Country

```
[14]: sales_performance=ds.groupby("Country").agg({
      "Revenue": "sum",
      "Quantity": "sum",
      "Invoice": "nunique"})
sales_performance=sales_performance.sort_values(by="Revenue",ascending=False) #Sorting Data in Descending order
sales_performance.columns=["Total Revenue","Total Quantity","Total Orders"] #Renaming Columns for better understanding
sales_performance=sales_performance.head(10)
print(sales_performance)
```

| Country        | Total Revenue | Total Quantity | Total Orders |
|----------------|---------------|----------------|--------------|
| United Kingdom | 8194777.533   | 4429046        | 26633        |
| EIRE           | 352242.730    | 188704         | 457          |
| Netherlands    | 263863.410    | 181823         | 150          |
| Germany        | 196290.351    | 107133         | 513          |
| France         | 130769.900    | 74471          | 297          |
| Sweden         | 51213.510     | 52238          | 83           |
| Denmark        | 46972.950     | 227030         | 33           |
| Switzerland    | 43343.410     | 22053          | 50           |
| Spain          | 37084.900     | 18332          | 85           |
| Australia      | 30051.800     | 20053          | 50           |

## Visualization of sales performance:

The result was then sorted by total revenue and lastly the performance was visualized using bar chart.

### Sales Performance Visualized by Bar Graph

```
[20]: top10=sales_performance.head(10)
plt.figure(figsize=(10,6))
plt.barh(top10.index, top10["Total Revenue"], color="skyblue")

plt.title("Top 10 Countries by Revenue")
plt.xlabel("Total Revenue")
plt.ylabel("Country")

plt.gca().invert_yaxis()

plt.tight_layout()
plt.ticklabel_format(style='plain', axis='x')
plt.show()
```



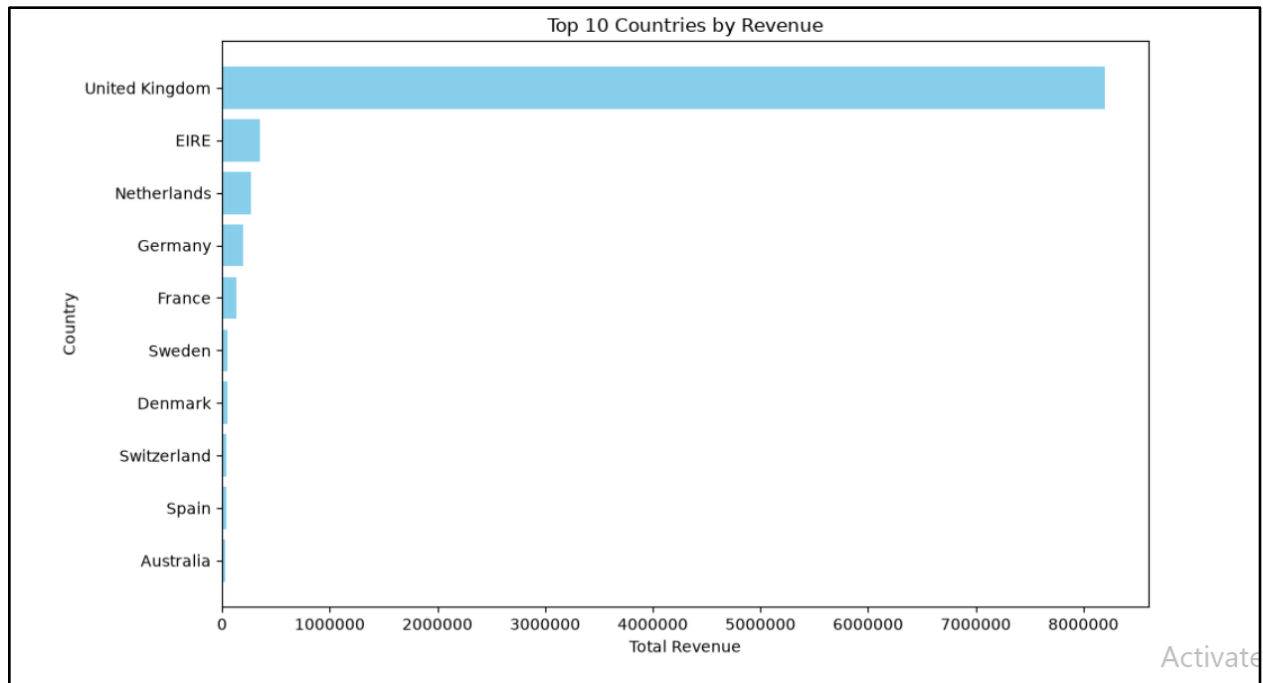


Figure 5: Country-wise sales performance

### Key Observations

According to fig.5, United Kingdom generated the highest revenue among all the other countries, contributing more than 8 million in total revenue. This indicates that the company has highest sales performance in UK as compared to other countries.

## Monthly Revenue Trend

- The invoice date column was converted into date time format and month was extracted from each column.
- The data was then grouped by month so total revenue generated per month can be calculated.
- A line graph was then plotted in order to understand the monthly trend.

### Task 8: Monthly Revenue Trend

```
[16]: ds["InvoiceDate"] = pd.to_datetime(ds["InvoiceDate"]) # Converting InvoiceDate to proper date format in python
      ds["Month"] = ds["InvoiceDate"].dt.to_period("M")
      monthly_revenue = ds.groupby("Month")["Revenue"].sum()

      plt.figure(figsize=(12,5))
      plt.plot(monthly_revenue.index.astype(str), monthly_revenue.values, label='Revenue')
      plt.legend()

      plt.title("Monthly Revenue Trend")
      plt.xlabel("Month")
      plt.ylabel("Revenue")
      plt.xticks(rotation=45)
      plt.tight_layout()
      plt.show()
```

## Line Graph

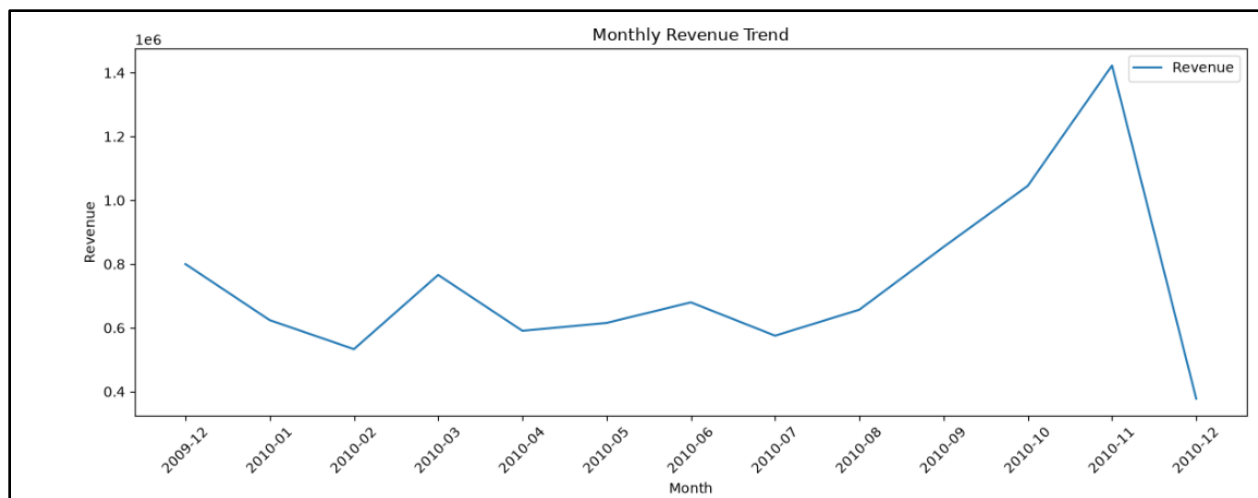


Figure 6: Monthly Revenue Trend

### Key Observations

According to the fig.6, the monthly trend fluctuated throughout the year. It keeps on increasing and decreasing during initial months but spiked towards the end of the year reaching its highest point in November 2010. Then there was a sharp decline in December 2010 which is likely due to missing or incomplete data present for the entire month.

## Correlation Heat map

Correlation heat map was used to examine the relation between Quantity, Price and Revenue. It was created using Seaborn in order to clearly visualize the correlation coefficients between these variables.

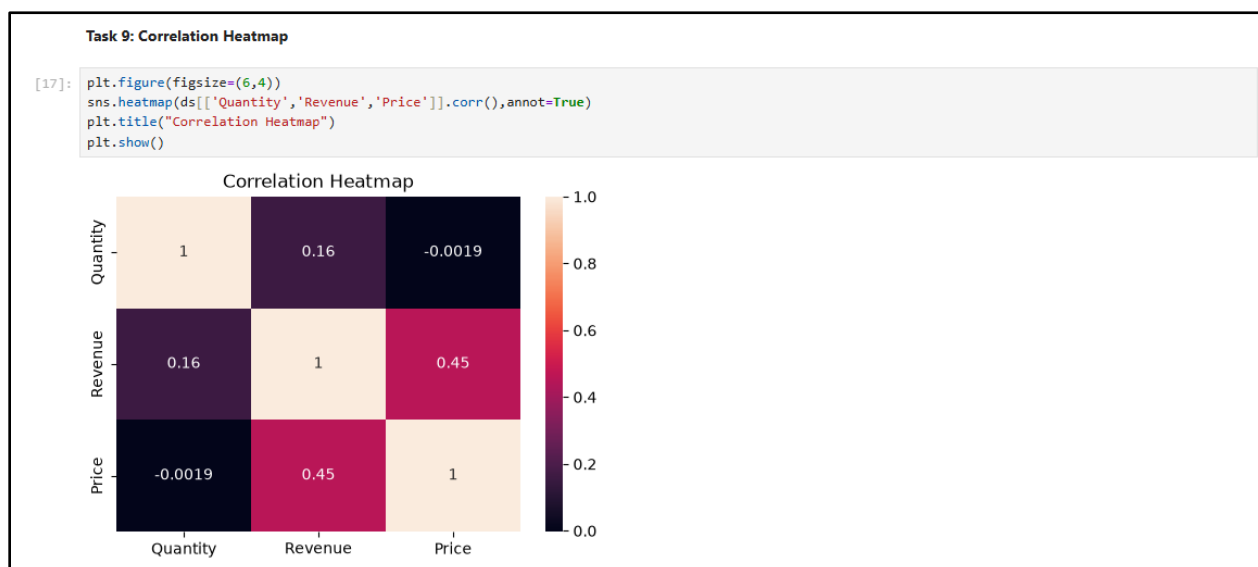


Figure 7: Correlation Heatmap

## Key Observations

Correlation value of 1 indicates a strong correlation between 2 variables. While a value closer to 0 indicates weak to moderate correlation.

From the fig.7 we can observe:

- Strong correlation exists when values are compared with themselves.
- Moderate positive correlation (**0.45**) is present between Revenue and Price indicating that higher priced product contributes to higher revenue.
- Weak positive correlation (**0.16**) exists between Quantity and Revenue indicating selling more products do not significantly affect the revenue.
- A weak or almost zero negative correlation (**-0.0019**) exists between Quantity and Price indicating that number of items purchased is not related to the price of the product.

## Outlier Detection Using Boxplot

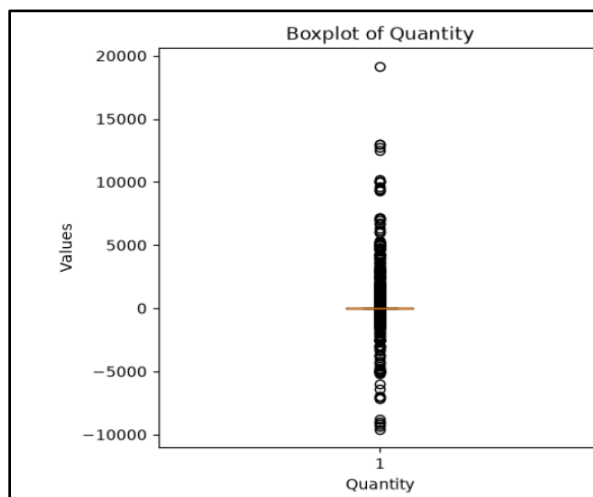
Using boxplot outlier were detected which are the extreme values present in the data that may influence the results of data analysis.

```
[24]: plt.figure(figsize=(5,5))
      plt.boxplot(ds['Quantity'])           # Outlier detection for Quantity
      plt.title("Boxplot of Quantity")
      plt.xlabel("Quantity")
      plt.ylabel("Values")
      plt.show()

      plt.figure(figsize=(5,5))
      plt.boxplot(ds['Price'])              # Outlier detection for Price
      plt.title("Boxplot of Price")
      plt.xlabel("Price")
      plt.ylabel("Values")
      plt.show()

      plt.figure(figsize=(5,5))
      plt.boxplot(ds['Revenue'])            # Outlier detection for Revenue
      plt.title("Boxplot of Revenue")
      plt.xlabel("Revenue")
      plt.ylabel("Values")
      plt.show()
```

### Boxplot for Quantity:

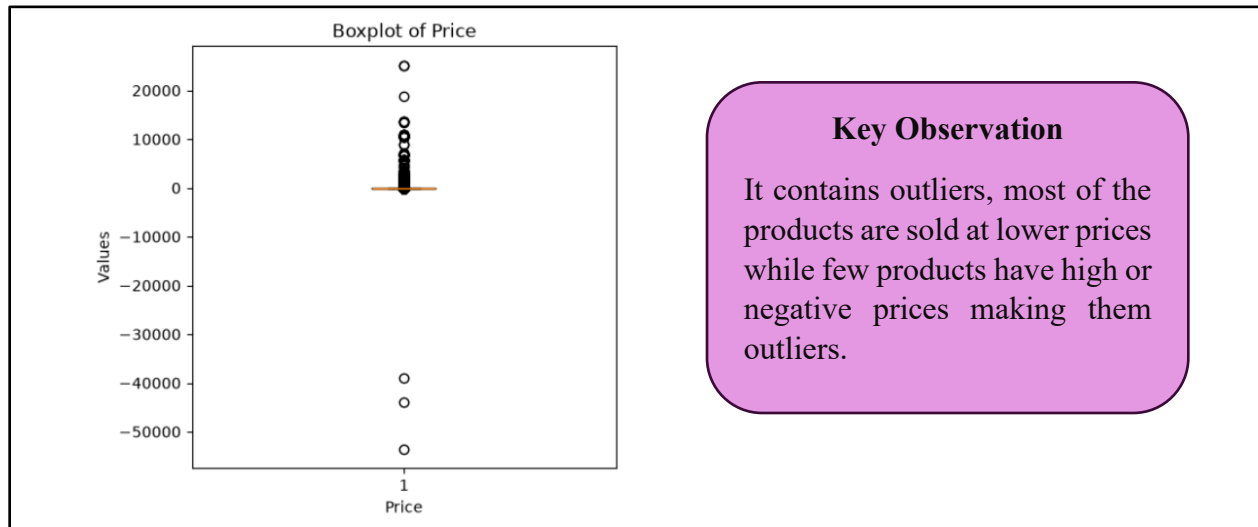


### Key Observation

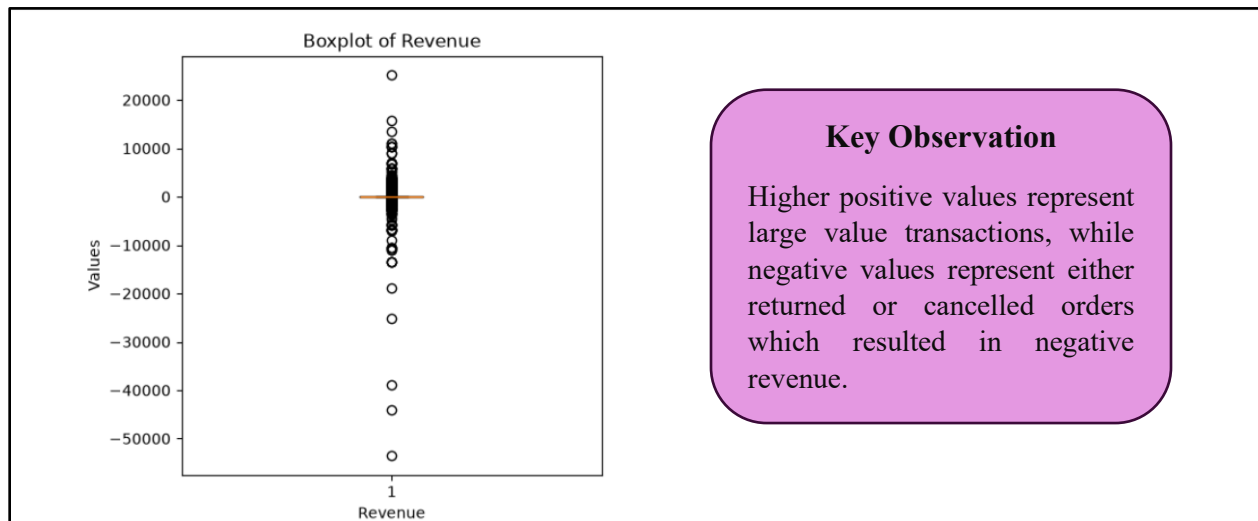
It shows several outliers above and below the normal range. The high positive values indicate bulk order while high negative values indicate returned or cancelled orders. While most of the transactions involve small quantities.

Figure 8: Quantity Boxplot

## Boxplot for Price:



## Boxplot for Revenue:



## Business Insights:

1. United Kingdom is the company's primary market as highest revenue was generated there as compared to other countries.
2. Monthly revenue fluctuated throughout the observed period which indicates that customer purchasing behaviors changes frequently.
3. Few products contributes more to the revenue and the total sales which indicates these products are in high demand.
4. Higher priced products contribute more to the revenue than the quantity sold.
5. Quantity boxplot indicates bulk purchases and Revenue boxplot indicates high-value transactions.

## Key Learnings

- Learned how to inspect read- world datasets.
- Performed data quality assessment.
- Created derived featured for analysis.
- Generated business insights through visualizations.
- Improved the understanding of EDA workflow using python.

## Conclusion

This first week of the AI/ML Internship focused on developing a strong foundation in data analysis and business understanding through Exploratory Data Analysis (EDA) of the Online Retail II dataset. By successfully examining the dataset, identifying data quality issues, analyzing customer purchasing behavior, evaluating sales performance across products and countries, and visualizing key trends, I gained practical experience in transforming raw data into meaningful business insights.

The task strengthened my understanding of essential data analysis techniques, including data inspection, handling missing values and duplicate records, revenue analysis, correlation analysis, trend visualization, and outlier detection using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn.

Overall, this week's experience enhanced both my analytical and technical skills while providing a solid foundation for more advanced data preprocessing, machine learning, and predictive analytics tasks in the upcoming weeks of the internship.